

Figure 1: Basic neuron design

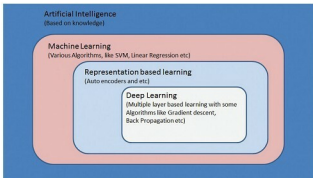


Figure 2: Layered learning mechanisms

field of study that gives computers the ability to learn without being explicitly programmed.” This was back in 1959.

In machine learning, computers are taught to solve certain problems with massive lists of rules, and are provided with models. In deep learning, the model provided can be evaluated with examples and a small set of instructions to modify it when it makes a mistake. Over time, a suitable model is able to solve the problem extremely accurately. That is why deep learning has become very popular and is catching every one’s eye. In the book ‘Fundamentals of Deep Learning’, the authors Nikhil Buduma and Nicholas Locascio state: “Deep learning is a subset of a more general field of artificial intelligence called machine learning, which is predicated on this idea of learning from example.”

Deep learning explained in detail

According to the free tutorial website *computer4everyone.com*, “Deep learning is a sub-field of machine learning concerned with algorithms inspired by the structure and function of the brain called artificial neural networks.”

Andrew Ng, who formally founded Google Brain, which eventually resulted in the commercialisation of deep learning technologies across a large number of Google services, has spoken and written a lot about deep learning. In his early talks on deep learning, Ng described

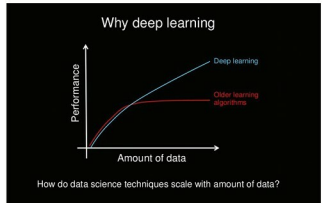


Figure 3: Why deep learning?

deep learning in the context of traditional artificial neural networks.

At this point, it would be unfair if I did not mention other experts who have contributed to the field of deep learning. They include:

- Geoffrey Hinton for restricted Boltzmann machines stacked as deep-belief networks (sometimes he is referred to as the father of machine learning)
- Yann LeCun for convolutional networks (he was a student of Hinton)
- Yoshua Bengio, whose team has developed Theano (an open source solution for deep learning)
- Juergen Schmidhuber, who developed recurrent nets and LSTMs (long short-term memory, which is a recurrent neural network (RNN) architecture)

According to Andrew Ng, because of the huge volumes of data now available for computation and the recent advances in algorithms, deep learning technology has been adopted across the globe pretty quickly.

The potential for applications of deep learning in the modern world is humongous. Application fields include speech synthesis, learning based on past history, facial recognition, self-driving cars, medical sciences, stock predictions, and real estate rate predictions, to name a few.

Prerequisites to understanding deep learning technologies

There are a number of discussion forums and blogs on whether one has to know deep mathematics to understand deep learning. In my view, this should be evaluated on a case-to-case basis. Having said that, it is better to know the following if you really want to understand deep learning and are serious about it:

- The basic functions of neural networks
- An understanding of the basics of calculus
- An understanding of matrices, vectors and linear algebra
- Algorithms (supervised, unsupervised, online, batch, etc)
- Python programming
- Case-to-case basis mathematical equations